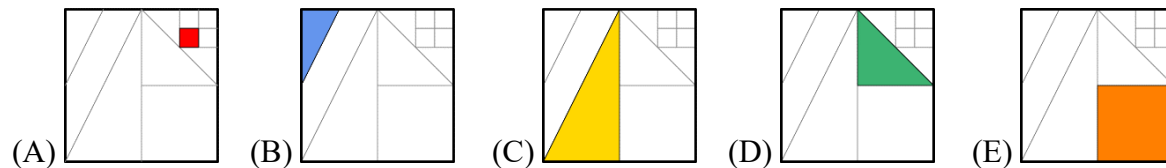


Math Kangaroo 2021 Sample Questions

LEVELS 5 AND 6

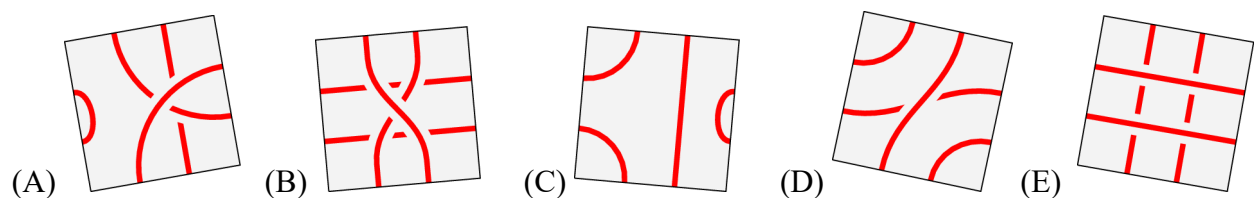
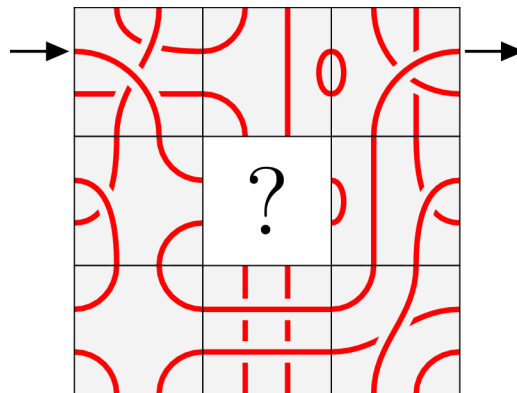
SAMPLE QUESTION FOR 3 POINTS

10. There is a square with line segments drawn inside it. The line segments are drawn either from the vertices or the midpoints of other line segments. We colored $\frac{1}{8}$ of the large square. Which one is our coloring?



SAMPLE QUESTION FOR 4 POINTS

13. Rosa wants to start at the arrow, follow the line, and get out at the other arrow. Which piece, if placed in the middle, cannot produce this? Note: The piece can be rotated.



SAMPLE QUESTION FOR 5 POINTS

29. 10 elves and trolls each were given a token with a different number from 1 to 10 written on it. They were each asked what number was on their token and all answered with a number from 1 to 10. The sum of the answers was 36. Each troll told a lie and each elf told the truth. What is the smallest number of trolls there could be in the group?

- (A) 1 (B) 3 (C) 4 (D) 5 (E) 7

Math Kangaroo 2021 Sample Questions

LEVELS 7 AND 8

SAMPLE QUESTION FOR 3 POINTS

5. When the five pieces shown fit together correctly, the result is a rectangle with a calculation written on it. What is the result of this calculation?

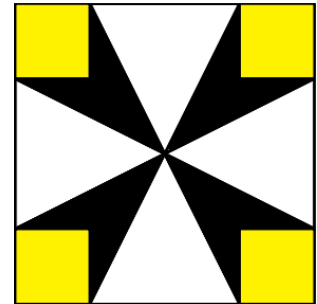


- (A) -100 (B) -8 (C) -1 (D) 199 (E) 208

SAMPLE QUESTION FOR 4 POINTS

13. The area of the large square is 16 cm^2 and the area of each small square is 1 cm^2 . What is the total area of the black flower?

- (A) 3 cm^2 (B) $\frac{7}{2} \text{ cm}^2$ (C) 4 cm^2 (D) $\frac{11}{2} \text{ cm}^2$ (E) 6 cm^2



SAMPLE QUESTION FOR 5 POINTS

26. 2021 colored kangaroos are arranged in a row and are numbered from 1 to 2021. Each kangaroo is colored red, gray, or blue. Among any three consecutive kangaroos, there are always kangaroos of all three colors. Bruce guesses the colors of five kangaroos. These are his guesses: Kangaroo 2 is gray; Kangaroo 20 is blue; Kangaroo 202 is red; Kangaroo 1002 is blue; Kangaroo 2021 is gray. Only one of his guesses is wrong. What is the number of the kangaroo whose color he guessed incorrectly?

- (A) 2 (B) 20 (C) 202 (D) 1002 (E) 2021